

Creation Date 29-Oct-2006

Revision Date 22-Mar-2024

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	Phosphorus(V) oxychloride
Cat No. :	R15200
Synonyms	Phosphoryl Chloride
Index No	015-009-00-5
CAS No	10025-87-3
EC No	233-046-7
Molecular Formula	Cl ₃ O P
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address

begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

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2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Substances/mixtures corrosive to metal

Category 1 (H290)

Health hazards

Acute oral toxicity

Category 4 (H302)

Acute Inhalation Toxicity - Vapors

Category 2 (H330)

Skin Corrosion/Irritation

Category 1 A (H314)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Specific target organ toxicity - (repeated exposure)

Category 1 (H372)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals

H302 - Harmful if swallowed

H330 - Fatal if inhaled

H314 - Causes severe skin burns and eye damage

H372 - Causes damage to organs through prolonged or repeated exposure

EUH014 - Reacts violently with water

EUH029 - Contact with water liberates toxic gas

Precautionary Statements

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor/physician

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

2.3. Other hazards

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Reacts violently with water
In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment
Lachrymator (substance which increases the flow of tears)
This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Phosphoric trichloride	10025-87-3	EEC No. 233-046-7	>95	Acute Tox. 4 (H302) Acute Tox. 2 (H330) Skin Corr. 1A (H314) Eye Dam. 1 (H318) STOT RE 1 (H372) Met. Corr. 1 (H290) EUH014 EUH029

REACH registration number

-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Inhalation	If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: After inhalation exposure, observe for 24 to 72 hours as pulmonary edema may be delayed

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4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Carbon dioxide (CO₂). Powder. CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

Contact with water liberates toxic gas. Water.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Contact with water liberates toxic gas. Reacts violently with water.

Hazardous Combustion Products

Oxides of phosphorus, Hydrogen chloride gas.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions

Should not be released into the environment.

6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Do not expose spill to water.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere.

Hygiene Measures

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Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Protect from moisture. Corrosives area. Keep under nitrogen. Keep away from water or moist air. Store under an inert atmosphere. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 6.1B

Switzerland - Storage of hazardous substances

Storage class - SC 6.1
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Phosphoric trichloride		STEL: 0.6 ppm 15 min STEL: 3.8 mg/m ³ 15 min TWA: 0.2 ppm 8 hr TWA: 1.3 mg/m ³ 8 hr	TWA / VME: 0.01 ppm (8 heures). indicative limit TWA / VME: 0.064 mg/m ³ (8 heures). indicative limit STEL / VLCT: 0.02 ppm. STEL / VLCT: 0.12 mg/m ³ .	TWA: 0.01 ppm 8 uren TWA: 0.064 mg/m ³ 8 uren STEL: 0.02 ppm 15 minuten STEL: 0.12 mg/m ³ 15 minuten	STEL / VLA-EC: 0.02 ppm (15 minutos). STEL / VLA-EC: 0.13 mg/m ³ (15 minutos). TWA / VLA-ED: 0.01 ppm (8 horas) TWA / VLA-ED: 0.064 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Phosphoric trichloride	TWA: 0.064 mg/m ³ 8 ore. Time Weighted Average TWA: 0.01 ppm 8 ore. Time Weighted Average STEL: 0.12 mg/m ³ 15 minuti. Short-term STEL: 0.02 ppm 15 minuti. Short-term	TWA: 0.02 ppm (8 Stunden). AGW - exposure factor 1 TWA: 0.13 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 0.02 ppm (8 Stunden). MAK TWA: 0.13 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.02 ppm Höhepunkt: 0.13 mg/m ³	STEL: 0.02 ppm 15 minutos STEL: 0.13 mg/m ³ 15 minutos TWA: 0.01 ppm 8 horas TWA: 0.064 mg/m ³ 8 horas	STEL: 0.12 mg/m ³ 15 minuten TWA: 0.064 mg/m ³ 8 uren	TWA: 0.01 ppm 8 tunteina TWA: 0.064 mg/m ³ 8 tunteina STEL: 0.02 ppm 15 minuutteina STEL: 0.13 mg/m ³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Phosphoric trichloride	MAK-KZGW: 0.02 ppm 15 Minuten MAK-KZGW: 0.12 mg/m ³ 15 Minuten MAK-TMW: 0.01 ppm 8 Stunden	TWA: 0.01 ppm 8 timer TWA: 0.064 mg/m ³ 8 timer STEL: 0.12 mg/m ³ 15 minutter STEL: 0.02 ppm 15	STEL: 0.04 ppm 15 Minuten STEL: 0.26 mg/m ³ 15 Minuten TWA: 0.02 ppm 8 Stunden	STEL: 0.12 mg/m ³ 15 minutach TWA: 0.064 mg/m ³ 8 godzinach	TWA: 0.064 mg/m ³ 8 timer TWA: 0.01 ppm 8 timer STEL: 0.13 mg/m ³ 15 minutter. value from the regulation

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	MAK-TMW: 0.064 mg/m ³ 8 Stunden	minutter	TWA: 0.13 mg/m ³ 8 Stunden		STEL: 0.02 ppm 15 minutter. value from the regulation
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Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Phosphoric trichloride	TWA: 0.064 mg/m ³ TWA: 0.01 ppm STEL : 0.12 mg/m ³ STEL : 0.02 ppm	TWA-GVI: 0.01 ppm 8 satima. TWA-GVI: 0.064 mg/m ³ 8 satima. STEL-KGVI: 0.02 ppm 15 minutama. STEL-KGVI: 0.13 mg/m ³ 15 minutama.	TWA: 0.01 ppm 8 hr. TWA: 0.064 mg/m ³ 8 hr. STEL: 0.02 ppm 15 min STEL: 0.12 mg/m ³ 15 min	STEL: 0.13 mg/m ³ STEL: 0.02 ppm TWA: 0.064 mg/m ³ TWA: 0.01 ppm	TWA: 0.06 mg/m ³ 8 hodinách. Ceiling: 0.12 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Phosphoric trichloride	TWA: 0.01 ppm 8 tundides. TWA: 0.064 mg/m ³ 8 tundides. STEL: 0.13 mg/m ³ 15 minutites. STEL: 0.02 ppm 15 minutites.		STEL: 0.02 ppm STEL: 0.13 mg/m ³ TWA: 0.01 ppm TWA: 0.064 mg/m ³	STEL: 0.12 mg/m ³ 15 percekben. CK TWA: 0.064 mg/m ³ 8 órában. AK	STEL: 0.02 ppm STEL: 0.12 mg/m ³ TWA: 0.01 ppm 8 klukkustundum. TWA: 0.064 mg/m ³ 8 klukkustundum. Ceiling: 0.2 ppm Ceiling: 1.2 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Phosphoric trichloride	STEL: 0.12 mg/m ³ STEL: 0.02 ppm TWA: 0.064 mg/m ³ TWA: 0.01 ppm	TWA: 0.064 mg/m ³ IPRD TWA: 0.01 ppm IPRD Oda STEL: 0.13 mg/m ³ STEL: 0.02 ppm	TWA: 0.064 mg/m ³ 8 Stunden TWA: 0.01 ppm 8 Stunden STEL: 0.12 mg/m ³ 15 Minuten STEL: 0.02 ppm 15 Minuten	TWA: 0.01 mg/m ³ TWA: 0.064 ppm STEL: 0.02 ppm 15 minuti STEL: 0.12 mg/m ³ 15 minuti	TWA: 0.15 ppm 8 ore TWA: 1 mg/m ³ 8 ore STEL: 0.8 ppm 15 minute STEL: 5 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Phosphoric trichloride	Skin notation MAC: 0.05 mg/m ³	TWA: 0.2 ppm TWA: 1.3 mg/m ³	TWA: 0.064 mg/m ³ 8 urah TWA: 0.01 ppm 8 urah STEL: 0.02 ppm 15 minutah STEL: 0.13 mg/m ³ 15 minutah	Binding STEL: 0.02 ppm 15 minuter Binding STEL: 0.13 mg/m ³ 15 minuter TLV: 0.01 ppm 8 timmar. NGV TLV: 0.064 mg/m ³ 8 timmar. NGV	

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See values below

Predicted No Effect Concentration (PNEC)

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Predicted No Effect Concentration (PNEC).

8.2. Exposure controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use only under a chemical fume hood.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles Face protection shield (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Butyl rubber	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Colorless
Odor	pungent
Odor Threshold	No data available
Melting Point/Range	1.2 °C / 34.2 °F
Softening Point	No data available
Boiling Point/Range	107 °C / 224.6 °F

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Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	1.11 mPa.s at 22 °C	
Water Solubility	Reacts violently with water	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Vapor Pressure	36 mbar @ 20 °C	
Density / Specific Gravity	1.645	
Bulk Density	Not applicable	Liquid
Vapor Density	5.3	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	Cl3 O P
Molecular Weight	153.33

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

Yes

10.2. Chemical stability

Reacts violently with water. Moisture sensitive. Contact with water liberates toxic gas.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	Reacts violently with water.

10.4. Conditions to avoid

Excess heat. Incompatible products. Exposure to moist air or water. Exposure to moisture.

10.5. Incompatible materials

Strong bases. Alcohols. Amines. Metals. Acids. Reducing Agent. Water. Oxidizing agent.

10.6. Hazardous decomposition products

Oxides of phosphorus. Hydrogen chloride gas.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;	
Oral	Category 4
Dermal	Based on available data, the classification criteria are not met
Inhalation	Category 2

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Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Phosphoric trichloride	LD50 = 380 mg/kg (Rat)	LD50 > 250 mg/kg (Rabbit)	LC50 = 308 mg/m ³ (Rat) 4 h

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Based on available data, the classification criteria are not met

(i) STOT-repeated exposure;

Category 1

Target Organs

Respiratory system.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Other Adverse Effects

Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. After inhalation exposure, observe for 24 to 72 hours as pulmonary edema may be delayed.

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Reacts with water so no ecotoxicity data for the substance is available.

12.2. Persistence and degradability

Persistence

Persistence is unlikely, based on information available.

Degradability

Decomposes in contact with water, Reacts with water.

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Degradation in sewage treatment plant

Decomposes in contact with water. Reacts violently with water.

12.3. Bioaccumulative potential

Product does not bioaccumulate due to reaction with water

12.4. Mobility in soil

Decomposes in contact with water. Reacts violently with water Is not likely mobile in the environment.

12.5. Results of PBT and vPvB assessment

Reacts violently with water. In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number

UN1810

14.2. UN proper shipping name

PHOSPHORUS OXYCHLORIDE

14.3. Transport hazard class(es)

6.1

Subsidiary Hazard Class

8

14.4. Packing group

I

ADR

14.1. UN number

UN1810

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14.2. UN proper shipping name	PHOSPHORUS OXYCHLORIDE
14.3. Transport hazard class(es)	6.1
Subsidiary Hazard Class	8
14.4. Packing group	I
IATA	FORBIDDEN FOR IATA TRANSPORT
14.1. UN number	UN1810
14.2. UN proper shipping name	PHOSPHORUS OXYCHLORIDE, FORBIDDEN FOR IATA TRANSPORT
14.3. Transport hazard class(es)	6.1
Subsidiary Hazard Class	8
14.4. Packing group	I
14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Phosphoric trichloride	10025-87-3	233-046-7	-	-	X	X	KE-28728	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Phosphoric trichloride	10025-87-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Phosphoric trichloride	10025-87-3	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Phosphoric trichloride	10025-87-3	Not applicable	Not applicable

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Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Phosphoric trichloride	WGK1	

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H290 - May be corrosive to metals
H302 - Harmful if swallowed
H330 - Fatal if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H372 - Causes damage to organs through prolonged or repeated exposure
EUH014 - Reacts violently with water
EUH029 - Contact with water liberates toxic gas

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

SAFETY DATA SHEET

Phosphorus(V) oxychloride

Revision Date 22-Mar-2024

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Prepared By

Health, Safety and Environmental Department

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22-Mar-2024

Revision Summary

New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

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End of Safety Data Sheet